

Serial Decode — user manual

Ian Ameline · version 1.11 · MIT licence

This app turns a Keithley bench instrument into a serial decoder. Clip onto a UART line, press **Capture**, and read the bytes on the front panel. You do not have to tell it the baud rate, the frame format or which way up the signal is — it works all of that out from the signal itself.

It was written on a **DMM6500** and that is **the only instrument it has been tested on**. It should run as it is on the rest of the Keithley TSP range that has a digitizer and the touchscreen app API — the DAQ6510 and DMM7510 in particular, and it will install on an SMU2461 and try — but nobody has run it there yet. **Which instruments** in the README says what is measured and what is inference.

Three things make it worth reaching for over a scope's decode: the **window is large** (a screenful is ~240 bytes, and a recording holds 32 768), every capture is **appended to the USB key automatically** while a key is in the slot, so you have a log without asking for one, and the **serial parameters are detected rather than configured**. For most debugging that is the whole job — connect, press, read.

Every number in this manual was measured on a real DMM6500 against a real signal generator unless the text says otherwise — and where something is calculated rather than measured, or untested, it says so in that spot. What that cannot cover is the variety of real devices, so if the app tells you something you can prove is wrong, that is the bug worth reporting. It is built so that **refusing with a reason is acceptable and confident nonsense is not**, and anything it is unsure about it says on the note row at the bottom of the screen.

Hooking it up — read this part

The line must stay between -10 V and +10 V. The app fixes the meter on its 10 V range. 3.3 V CMOS, 5 V TTL and a 6 V LIN line tapped to ground are all fine as they are. For a 12 V line, put a 2:1 resistor divider in front of it — and if it is an open-drain bus, pick resistors big enough not to load it down.

Connect INPUT LO to the ground of the circuit you are probing. Everything is measured at INPUT HI relative to INPUT LO, so without a shared ground the readings mean nothing.

Do not put a differential bus straight across HI and LO. RS-485, CAN and LIN need to go through a transceiver first, or you need to tap one side of the pair against ground.

A DC offset does not matter. The app measures the two voltage levels on your line and puts its decision threshold between them, so it does not care where they sit — only that both fit inside ± 10 V. A 1.6 V swing sitting up at 6 V decodes fine. **The tested range is a 0.33 V swing to an 8 V swing**, and signals spanning 9.3 V peak-to-peak decode when the extra span is spikes on a smaller logic swing. A 60 mV swing is refused outright.

A small swing needs a quiet line, not a special setting. The noise the app tolerates is a fraction of your swing, around 40 %, so 50 mV of noise is nothing on a 3.3 V line and a tenth of the budget on a 0.5 V one.

Quick start

1. Connect the line to **INPUT HI** and its ground to **INPUT LO**.

2. Press **Capture** while the device is talking.
3. Read the bytes. Press **View** to switch between hex and plain text.

That is the whole normal workflow. The first capture also locks the baud rate when it is confident, so later captures are faster.

Two more things worth knowing before you need them: **Mode** turns Capture into a recording of up to 32 768 bytes straight to the USB key, still one press — and once a long job starts, **nothing stops it early**, so decide the size before you press — see **The TRIGGER key** below.

The screen

SERIAL DECODE - 240B FRAME

End App

BAUD	FORMAT	IDLE	LOGIC	THRESH	SA/BIT	RATE	BYTES	ERR	FIT	S/N
9600	8N1	HIGH	3V3 CMOS	1.63 V	8.3	80 kS/s	239	0	0.98	79 dB
Start bit auto-locked 9600 baud 8N1: next 80 kS/s ~240 B (Options > Baud Rate = 0 re-detects)										
0000	81 75 74 20 6C 61 62 6F	72 65 20 65 74 20 64 6F	?ut labore et do							
0016	6C 6F 72 65 20 6D 61 67	6E 61 20 61 6C 69 71 75	lore magna aliqu							
0032	61 2E 20 55 74 20 65 6E	69 6D 20 0D 0A 61 64 20	a. Ut enim ..ad							
0048	6D 69 6E 69 6D 20 76 65	6E 69 61 6D 2C 20 71 75	minim veniam, qu							
0064	69 73 20 6E 6F 73 74 72	75 64 20 65 78 65 72 63	is nostrud exerc							
0080	69 74 61 74 69 6F 6E 20	75 6C 6C 61 6D 63 6F 20	itation ullamco							
0096	6C 61 62 6F 72 69 73 20	6E 69 73 69 20 0D 0A 75	laboris nisi ..u							
0112	74 20 61 6C 69 71 75 69	70 20 65 78 20 65 61 20	t aliquip ex ea							
0128	63 6F 6D 6D 6F 64 6F 20	63 6F 6E 73 65 71 75 61	commodo consequa							
0144	74 2E 20 44 75 69 73 20	61 75 74 65 20 69 72 75	t. Duis aute iru							
0160	72 65 20 64 6F 6C 6F 72	20 69 6E 20 72 65 70 72	re dolor in repr							
0176	65 68 65 6E 64 65 72 69	74 20 0D 0A 69 6E 20 76	ehenderit ..in v							
0192	6F 6C 75 70 74 61 74 65	20 76 65 6C 69 74 20 65	oluptate velit e							
0208	73 73 65 20 63 69 6C 6C	75 6D 20 64 6F 6C 6F 72	sse cillum dolor							
0224	65 20 65 75 20 66 75 67	69 61 74 20 6E 75 6C	e eu fugiat nul							

FRAME HEX pg 1/1 bytes 1-239/239 win 240 [done]

log: bytes568 2968 B

Capture

View

Mode

NewLog

Save

Options

Along the top are the things the app worked out about your line:

Field	Meaning
BAUD	The rate it found. A padlock means it is locked and will not be re-detected.
FORMAT	Data bits, parity, stop bits — 8N1 , 7E1 and so on.
IDLE	What the line rests at: HIGH for CMOS/TTL, LOW for an RS-232-style line.
LOGIC	The family it looks like, e.g. 3V3 CMOS , 5V TTL .
THRESH	The voltage it is using to tell 0 from 1.
SA/BIT	Samples per bit. More is better; 8 is comfortable, 4 still works.
RATE	How fast the meter sampled.
BYTES / ERR	How many bytes came out, and how many of them you cannot trust.

Field	Meaning
FIT	How well one bit-length explains what it measured, from 0 to 1. Below about 0.9 means the timing was messy.
S/N	How far your two logic levels stand above the noise sitting on them, in dB. Bigger is quieter.

A dash means "not measured yet".

FIT and **S/N** are coloured, and the colour is the quick read. Green means there is margin, amber means there is some but not much, red means you are close to the edge — so the top line can be checked at a glance instead of remembering two sets of numbers:

	green	amber	red
FIT	0.90 and up	0.72 to 0.89	below 0.72
S/N	30 dB and up	15 to 29 dB	below 15 dB

FIT 0.90 is the same figure auto-lock requires, so a green **FIT** is also "this capture was good enough to lock". **Red means close to failing, not failing.** A red **S/N** still decodes: the point where captures stop working is around 12 dB, and past it the app says `no clear logic levels` rather than handing you wrong bytes. Green **S/N** is noise under about 5 % of your swing, amber down to about a third of it.

S/N describes your wiring, not the app. It is measured on the flat parts of the signal, away from the edges, so it reports what is riding on your levels — mains hum picked up by a long unshielded lead, a switching supply, a noisy ground. If it reads low, the wiring is worth fixing before anything else: every other tolerance in this manual is a fraction of the swing, so noise spends the same budget that jitter and slow edges do. The cell stops at `>99 dB` rather than counting higher: above that it would be reporting the meter's own digitizer resolution rather than anything about your line, so what it shows there is a limit and not a measurement.

Broken bytes are shown, not hidden. **ERR** turns red when there are any, and every row containing one is drawn red across its whole width — offset, hex and ASCII together — so damage is found by looking rather than by counting. Flagged bytes read as `?` in the ASCII gutter:

SERIAL DECODE - 240B FRAME

End App

BAUD	FORMAT	IDLE	LOGIC	THRESH	SA/BIT	RATE	BYTES	ERR	FIT	S/N
9600	8N1	HIGH	3V3 CMOS	1.63 V	8.3	80 kS/s	238	5	1.00	79 dB

Start bit

began mid-byte -- the first 5 bytes are misaligned.

0000	A5 8D A5 85 81 64 65 73	65 72 75 6E 74 20 6D 6F	?deserunt mo
0016	6C 6C 69 74 20 0D 0A 61	6E 69 6D 20 69 64 20 65	llit ..anim id e
0032	73 74 20 6C 61 62 6F 72	75 6D 2E 20 4C 6F 72 65	st laborum. Lore
0048	6D 20 69 70 73 75 6D 20	64 6F 6C 6F 72 20 73 69	m ipsum dolor si
0064	74 20 61 6D 65 74 2C 20	63 6F 6E 73 65 63 74 65	t amet, consecte
0080	74 75 72 20 61 64 69 70	69 73 63 69 6E 67 20 0D	tur adipiscing .
0096	0A 65 6C 69 74 2C 20 73	65 64 20 64 6F 20 65 69	.elit, sed do ei
0112	75 73 6D 6F 64 20 74 65	6D 70 6F 72 20 69 6E 63	usmod tempor inc
0128	69 64 69 64 75 6E 74 20	75 74 20 6C 61 62 6F 72	ididunt ut labor
0144	65 20 65 74 20 64 4C 6F	72 65 6D 20 69 70 73 75	e et dLorem ipsu
0160	6D 20 64 6F 6C 6F 72 20	73 69 74 20 61 6D 65 74	m dolor sit amet
0176	2C 20 63 6F 6E 73 65 63	74 65 74 75 72 20 61 64	, consectetur ad
0192	69 70 69 73 63 69 6E 67	20 65 6C 69 74 2C 20 73	ipiscing elit, s
0208	65 64 20 64 6F 20 0D 0A	65 69 75 73 6D 6F 64 20	ed do ..eiusmod
0224	74 65 6D 70 6F 72 20 69	6E 63 69 64 69 64	tempor incidid

FRAME HEX pg 1/1 bytes 1-238/238 win 240 [done]

log: bytes568 3921 B

Capture

View

Mode

NewLog

Save

Options

That is a real 240B capture that began part way through a byte. **The note row names the cause** — began mid-byte -- the first N bytes are misaligned. — which is the difference between a puzzle and a known condition: the app arms on a busy line, so it lands wherever the traffic happens to be, and the bytes before the first gap between messages cannot be framed. On a busy line roughly one capture in eight starts this way.

The note and ERR agree, and that is deliberate. The note's number is how far the damage reaches: the last byte in the opening region the decoder could not frame, so "the first N" is literally true and nothing past N is being accused. ERR is then never less than N. Above, the note says 5 and ERR says 5. ERR counts the whole region rather than only the frames that failed a check, because inside a misaligned head the bit boundaries are in the wrong places: a frame there can satisfy its parity and its stop bit by luck and still hand back a perfectly ordinary character that is the wrong byte. **ERR counts bytes you cannot trust, not checks that failed**, because the silently wrong ones are the dangerous ones.

Two exclusions remain, and both keep ERR at 0 on good data. When no misaligned head is identified, the first three frames are ignored — a gapless line has no idle for the framer to anchor on, so it routinely resynchronises a frame or two in, and counting that made ERR read 1 or 2 on every healthy capture. The final frame is ignored always, because the capture boundary cuts it in half. Neither exclusion applies across a head the app has evidence for, which is what stopped the panel reading ERR 1 beside a note naming four bytes.

Everything from the first gap onward decoded cleanly, and the log on the USB key holds every byte: one that failed its parity or stop bit is still written, marked, rather than dropped.

Underneath: the **trigger cell**, the bytes, then two rows of text. The **status row** tells you where you are (FRAME HEX pg 1/1 bytes 1-155/155 win 240 [done]) and names the log file.

Which mode you are in is in the title bar — SERIAL DECODE - 240B FRAME , - 8K CAPTURE or - 32K CAPTURE , in the largest type on the screen.

The cell at the left of the note row is what a capture **WAITS for**, colour-coded so a glance is enough: green **Start bit** (the normal case — it waits for the line's own start bit), yellow **Free run** (it does not wait for anything), cyan **Trigger key** (it waits for the physical key).

The cell at the bottom right is the rear **BNC**, and it is **empty when the rear BNC is off** — which is the default. When it is not, it names the direction: green **EXT TRIG IN**, amber **EXT TRIG OUT**, cyan **EXT TRIG I/O**. A trailing **?** means the input is switched on but *not in force*, which happens under **Free run** — free run means do not wait, so there is nothing for the input to gate.

During a recording the status row carries a live counter — **recording... 40 % of the buffer**, then **decoding... 4096/8192 bytes** — with a **cyan progress bar** beside it, and the cell to its right reads **Capture or Mode = Stop**.

View switches the byte display to plain text, which is what you want for anything human-readable — and a row carrying a flagged byte is red here too:

SERIAL DECODE - 240B FRAME

End App

BAUD	FORMAT	IDLE	LOGIC	THRESH	SA/BIT	RATE	BYTES	ERR	FIT	S/N
9600	8N1	HIGH	3V3 CMOS	1.63 V	8.3	80 kS/s	237	1	1.00	76 dB
Start bit	began mid-byte -- the first byte is misaligned.									
? Excepteur sint occaecat ..cupidatat non proident, sunt in culpa qui officia de serunt mollit ..anim id est laborum. Lorem ipsum dolor sit amet, consectetur adi piscing ..elit, sed do eiusmod tempor incididunt ut labore et dLorem ipsum do										

FRAME TEXT pg 1/1 bytes 1-237/237 win 239 [done]log: bytes568 3205 B

CaptureViewModeNewLogSaveOptions

The note row is the important one. Warnings appear there — an ambiguous baud rate, a line that disagrees with the rate you locked, a recording that stopped early. If more than one applies it ends with **(+2 more)**, and pressing **Save** writes all of them to a file.

Here it is doing that — a capture that started part way through a byte, which the app detected and said so rather than showing the misaligned bytes without comment — four of them, here:

SERIAL DECODE - 240B FRAME

End App

BAUD	FORMAT	IDLE	LOGIC	THRESH	SA/BIT	RATE	BYTES	ERR	FIT	S/N
9600	8N1	HIGH	3V3 CMOS	1.63 V	8.3	80 kS/s	236	4	1.00	80 dB

Start bit began mid-byte -- the first 4 bytes are misaligned.

0000	B1 A5 D1 B1 20 73 65 64	20 64 6F 20 65 69 75 73	...? sed do eius
0016	6D 6F 64 20 74 65 6D 70	6F 72 20 69 6E 63 69 64	mod tempor incid
0032	69 64 75 6E 74 20 75 74	20 6C 61 62 6F 72 65 20	idunt ut labore
0048	65 74 20 64 4C 6F 72 65	6D 20 69 70 73 75 6D 20	et dLorem ipsum
0064	64 6F 6C 6F 72 20 73 69	74 20 61 6D 65 74 2C 20	dolor sit amet,
0080	63 6F 6E 73 65 63 74 65	74 75 72 20 61 64 69 70	consectetur adip
0096	69 73 63 69 6E 67 20 65	6C 69 74 2C 20 73 65 64	iscing elit, sed
0112	20 64 6F 20 0D 0A 65 69	75 73 6D 6F 64 20 74 65	do ..eiusmod te
0128	6D 70 6F 72 20 69 6E 63	69 64 69 64 75 6E 74 20	mpor incididunt
0144	75 74 20 6C 61 62 6F 72	65 20 65 74 20 64 6F 6C	ut labore et dol
0160	6F 72 65 20 6D 61 67 6E	61 20 61 6C 69 71 75 61	ore magna aliqua
0176	2E 20 55 74 20 65 6E 69	6D 20 0D 0A 61 64 20 6D	. Ut enim ..ad m
0192	69 6E 69 6D 20 76 65 6E	69 61 6D 2C 20 71 75 69	inim veniam, qui
0208	73 20 6E 6F 73 74 72 75	64 20 65 78 65 72 63 69	s nostrud exerci
0224	74 61 74 69 6F 6E 20 75	6C 6C 61 6D	tation ullam

FRAME HEX pg 1/1 bytes 1-236/236 win 240 [done]

log: bytes568 4877 B

Capture

View

Mode

NewLog

Save

Options

An empty note row is not a promise that everything is perfect. It means the app did not detect anything wrong. A noise spike landing inside a data bit changes that byte with nothing to give it away.

The buttons

Button	Does
Capture	Take a capture and decode it. In a recording mode, does the whole recording.
View	Switch the display: hex + ASCII → plain text → back.
Mode	Choose what Capture does: a screenful, an 8 kB recording, or a 32 kB one.
NewLog	Start a new numbered log file on the USB key.
Save	Write a full report of what is on screen to the USB key.
Options	The settings screen.
Up / Dn	Right-hand edge. Only appear when there is more than one page. Once you press either, page N of M appears at the right of the note row — and stays for later captures, which is why some figures here show it on a screen nobody has just paged.
Lock Rate	Right-hand edge. Only appears when the rate is not locked.

Every button responds immediately except the ones that take a capture. **Capture takes about 1.7 seconds when the rate is locked, and 4 to 9 seconds when it is not** — because to find an unknown rate it has to capture two or three times. Slower lines take longer, not less. The exact time varies from press to press because it depends on where the traffic happens to be when you press.

A press during a capture is not lost — it is **queued** and acted on when the capture finishes. So if the panel seems to ignore you, wait rather than pressing again.

Paging through a long capture

SERIAL DECODE - 32K CAPTURE
End App

BAUD	FORMAT	IDLE	LOGIC	THRESH	SA/BIT	RATE	BYTES	ERR	FIT	S/N
9600	8N1	HIGH	3V3 CMOS	1.63 V	4.1	39 kS/s	32768	1	1.00	35 dB
Start bit panel shows bytes 24577-32768 of 32768; all are in bytes569.txt page 4 of 35										
25296	6D 2C 20 71 75 69 73 20	6E 6F 73 74 72 75 64 20	m, quis nostrud exercitation ull amco laboris nis i ..ut aliquip e x ea commodo con sequat. Duis aut e irure dolor in reprehenderit . .in voluptate ve lit esse cillum dolore eu fugiat nulla pariatur. ..Excepteur sin t occaecat cupid atat non proiden Up Dn							
25312	65 78 65 72 63 69 74 61	74 69 6F 6E 20 75 6C 6C								
25328	61 6D 63 6F 20 6C 61 62	6F 72 69 73 20 6E 69 73								
25344	69 20 0D 0A 75 74 20 61	6C 69 71 75 69 70 20 65								
25360	78 20 65 61 20 63 6F 6D	6D 6F 64 6F 20 63 6F 6E								
25376	73 65 71 75 61 74 2E 20	44 75 69 73 20 61 75 74								
25392	65 20 69 72 75 72 65 20	64 6F 6C 6F 72 20 69 6E								
25408	20 72 65 70 72 65 68 65	6E 64 65 72 69 74 20 0D								
25424	0A 69 6E 20 76 6F 6C 75	70 74 61 74 65 20 76 65								
25440	6C 69 74 20 65 73 73 65	20 63 69 6C 6C 75 6D 20								
25456	64 6F 6C 6F 72 65 20 65	75 20 66 75 67 69 61 74								
25472	20 6E 75 6C 6C 61 20 70	61 72 69 61 74 75 72 2E								
25488	20 0D 0A 45 78 63 65 70	74 65 75 72 20 73 69 6E								
25504	74 20 6F 63 63 61 65 63	61 74 20 63 75 70 69 64								
25520	61 74 61 74 20 6E 6F 6E	20 70 72 6F 69 64 65 6E								
32 kB 32768 bytes 1 err 68 win -> bytes569.txt [cap] log: on first capture										

Capture

View

Mode

NewLog

Save

Options

A recording is far more than one screen, so **Up** and **Dn** appear down the right-hand edge and **page N of M** appears at the right of the note row — the two numbers in white, the words dimmed, so the count reads at a glance. The note beside it says which bytes are on screen and that the whole run is in the file, which is the thing you cannot otherwise tell from a screenful taken out of the middle:

panel shows bytes 24577-32768 of 32768; all are in bytes569.txt

The offset column agrees with it — the rows start at 25296, not at zero — and the status row carries the run's own summary: 32768 bytes, 1 it cannot vouch for, 68 windows, and the file it went to. **That 1 is the same figure as ERR**, not a second opinion about it.

You page through the tail, not the whole run. The panel keeps the last **8 192 bytes** of a recording however long the run was; the file keeps all of it. That is why this 32 768-byte capture shows bytes 24577-32768 and pages 35, not 137 — and why the count is the same in either recording mode, since 8 kB is its own tail.

The page count follows the view. Press **View** on that same capture and it becomes 7 pages instead of 35, because a text row holds 80 characters and a hex row holds 16 bytes — 1200 bytes a page against 240, over the same 8 192:

SERIAL DECODE - 32K CAPTURE

End App

BAUD	FORMAT	IDLE	LOGIC	THRESH	SA/BIT	RATE	BYTES	ERR	FIT	S/N
9600	8N1	HIGH	3V3 CMOS	1.63 V	4.1	39 kS/s	32768	1	1.00	35 dB

Start bit

panel shows bytes 24577-32768 of 32768; all are in bytes569.txt

page 7 of 7

Up

Dn

32 kB 32768 bytes 1 err 68 win -> bytes569.txt [cap]

log: on first capture

Capture

View

Mode

NewLog

Save

Options

Nothing about the capture changed; only how much of it fits on a screen. The byte range in the note row is the same, and so is the summary.

The TRIGGER key — the one hardware button that matters

The **TRIGGER** key is the physical key on the front panel, to the **right** of the screen — the lowest of that column of buttons. It is not part of the app's own row of buttons. It was intended to do two things nothing on the glass can do; **one of them works** — timing a capture — and the other, stopping a long job, does not. Both are described below, because knowing which is which is the difference between waiting confidently and pressing a key that does nothing.

1. It does not stop a long job, and nothing else does either

A recording runs to its own end. Once you press **Capture** in `8 kB` or `32 kB`, nothing on the instrument will cut it short — not a button on the glass, and not the TRIGGER key. Decide before you press. The status row says so before you commit:

```
32 kB  ready -- 32768 bytes to a file, no stop once started
```

There is deliberately no time estimate. Two different durations apply and only one is knowable here: the time your *device* takes to send that many bytes, and the meter's decode, which costs several times more — 8 kB is a couple of seconds on the wire and around half a minute to decode. A single number would be reassuring and wrong, so what you get is the exact byte ceiling and a progress bar.

What does bound a run, so it can never hold the panel indefinitely:

- the byte ceiling for the mode — 8192 or 32768;
- a quiet line: one second with no activity ends it;

- under flow control, 32 windows or 20 minutes of wall clock, whichever comes first. The 20 minutes is what matters on slow lines, where a full 32 kB window would take 18 minutes at 300 baud.

A tap on Capture during a run is not a stop. It is queued by the instrument and delivered when the run finishes. The app recognises presses made during the run it has just finished and discards them, saying so on the note row, rather than starting another recording you did not ask for.

2. It can time a capture

Set `Options ▶ Trigger = Trigger key`, press **Capture**, and the capture waits for you: it fires when you press the physical key, so you decide the moment the line is sampled. Verified on this hardware — armed on a deliberately idle line, the capture fired on the key press rather than timing out.

If the trigger you chose never arrives, the capture goes ahead free-running after a few seconds rather than failing, and the note row says which one was missing.

So the key has exactly one job here: it times a capture, and only when you have asked for that in `Options ▶ Trigger`. At every other moment — including all the way through a recording — it does nothing.

How much you can capture at once

Four answers, and almost everyone only ever needs the first. **All of them are one press.**

You need	Use	Get
a look at what a device is saying	FRAME — just press Capture	~ 240 bytes , on screen
a longer burst, quickly	Mode → <code>8 kB</code> , then Capture	up to 8 192 bytes , to a file
the longest single recording	Mode → <code>32 kB</code> , then Capture	up to 32 768 bytes , to a file
more than that, losslessly	flow control, and a device that obeys it	as much as you like, ~20 minutes per press

FRAME is the normal case and covers almost all debugging. One press gives you a screenful, already decoded, with the baud rate and frame format worked out for you and a copy appended to the USB key automatically. You do not have to set anything up or lock anything down.

Recording to a file

Press **Mode** until the cell reads `8 kB` or `32 kB`. **You must lock the baud rate first** — press **Lock Rate**, or type the rate into **Options**. Then **press Capture once**, and that press does the whole job: it records, decodes every byte, writes them to the USB key, and comes back with the tail on screen. There is nothing to press in the middle.

SERIAL DECODE - 8K CAPTURE

End App

BAUD	FORMAT	IDLE	LOGIC	THRESH	SA/BIT	RATE	BYTES	ERR	FIT	S/N
9600	8N1	--	?	--	--	40 kS/s	--	--	0.00	--
Start bit										

8 kB recording... 98 % of the buffer

no stop until it ends

Capture

View

Mode

NewLog

Save

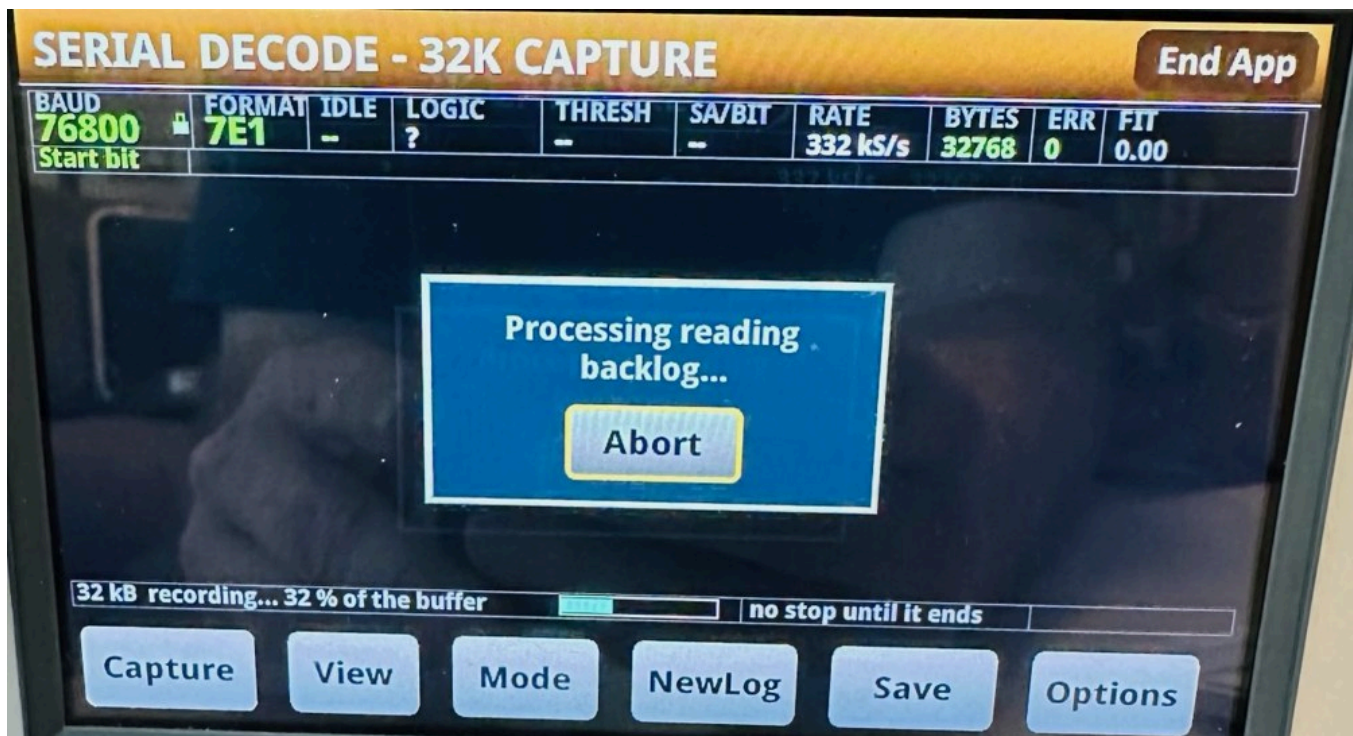
Options

While it works, the counter on the status row moves — first `recording... N % of the buffer`, then `decoding... N/8192 bytes` — with a cyan progress bar beside it, and the cell to its right says whether anything can stop it. The screenshot above is a real 8 kB run caught in the first of those two phases, at 98 % of the buffer, with `no stop until it ends` in that cell; the top row reads `--` across because nothing has been decoded yet to describe, the dump area is blank for the same reason, and the note row is empty because there is nothing wrong. **There is no way to finish it early** — see the section on the TRIGGER key above for what bounds it instead.

When it finishes, the note row goes quiet. The status row's summary — bytes, errors, window count and filename — is the completion message; the yellow note row is reserved for what went wrong, so an empty one after a recording means there was nothing to say.

It stops by itself when the window is full or when the line has been quiet for a second. There is also a 20-minute backstop, so nothing can wait for ever — it only matters on very slow lines, where a full 32 kB window takes 18 minutes at 300 baud.

Nothing is dropped inside a recording. What you get is a continuous, in-order run of the line — verified with a 1024-byte non-repeating test pattern, where the decoded bytes were one unbroken slice of it.



If a **Processing reading backlog...** dialog appears, ignore it. Do not press Abort.

It is the instrument's own message, not the app's: the digitizer is producing readings faster than the meter is moving them out of the buffer. It **clears by itself** and the recording carries on normally.

Seen on the bench from 38.4 kBd upwards — at 182 kS/s, at 635 kS/s recording a 76.8 kBd line, and within a second of pressing Capture at 1 MS/s on a 115.2 kBd line. Every one of those recordings completed and decoded correctly, including a full 32 kB one.

Pressing **Abort** would stop the acquisition the dialog is describing, which is the one thing that actually loses the capture. There is nothing to do but let it run.

The photograph above is that dialog over a live 32 kB recording of a 76.8 kBd line at 332 kS/s, with the progress bar behind it at 32 % of the buffer and the row beside it reading **no stop until it ends**. It is a photograph rather than a screenshot for a reason worth knowing: **the instrument's web interface stops answering while the digitizer runs**, so the remote screen grab cannot capture this dialog at all -- eight attempts timed out. If you need to document or report something that happens during an acquisition, a camera is the only way to see it.

Which window size to pick

The window is the unit of work: one press records a window, decodes it, and files it. So the size is a straight trade, and neither choice affects what gets decoded — both are lossless.

	8 kB	32 kB
Bytes per press	8 192	32 768
Recording, at 9600 baud	~8.5 s	~34 s
Decoding afterwards	~30 s	~2 minutes
Best for	seeing something soon, and stopping soon	the most data for the fewest presses

Decoding is the slow part, and it is slower than the line: the meter decodes roughly 300 bytes a second whatever the baud rate. That is why **8 kB** feels much more responsive even though both are one press.

The app starts in FRAME, and one press of **Mode** reaches **8 kB**, a second **32 kB**. Stop at **8 kB** when you would rather see bytes soon than see all of them, or while you are still working out whether you are probing the right pin; go on to **32 kB** when you want the most data for the fewest presses.

SERIAL DECODE - 32K CAPTUREEnd App

BAUD	FORMAT	IDLE	LOGIC	THRESH	SA/BIT	RATE	BYTES	ERR	FIT	S/N
9600	8N1	HIGH	3V3 CMOS	1.63 V	4.1	39 kS/s	8192	1	1.00	56 dB
Start bit	page 1 of 7									

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UpDn

32 kB ready – 32768 bytes to a file, no stop once startedlog: on first capture

CaptureViewModeNewLogSaveOptions

Above: an 8 kB recording has finished and **Mode** has been pressed once, so the title bar reads **32K CAPTURE** while the previous run's retained tail is still on screen. Note what the panel is saying — **Up** and **Dn** have appeared because the tail is more than one page, the status row states the next run's ceiling and that nothing will stop it, and **the note row is empty**, because a recording that worked has nothing to warn about.

The mode you choose stays chosen. The title bar says which one you are in — **SERIAL DECODE – 240B FRAME**, **– 8K CAPTURE** or **– 32K CAPTURE** — and a recording that finishes leaves it alone, so pressing **Capture** again records another window. Only **Mode** changes it. Two things move it for you, and both say so on the note line: **Options ▶ Auto Detect**, because auto-detection cannot run in a recording mode at all, and clearing the baud rate, because a recording needs it locked.

More than one window: flow control

A window is not a total. If your device is built to wait, you can capture as much as you like without losing a byte, **and without pressing anything per window** — up to 32 windows or 20 minutes per press, whichever comes first. To carry on past that, press **Capture** again; it resumes where it stopped, in a new file.

Set **Options ▶ Rear BNC = FC Out**. The rear EXT TRIG OUT then emits **one pulse each time a capture is armed** (measured on a scope: idle-low at +0.20 V, +4.45 V high, 5.72 µs wide with a 436 ns rise — the width is the meter's own default and this firmware does not let the app change it) — a credit meaning "send now, and send no more than one window's worth". Then one press of **Capture** runs the whole conversation: credit, record, decode, file, credit again — round and round, with the window number on the

status row, until the device stops sending. While the app is decoding, nothing is armed, no credit is issued, and a device that waits stays quiet, so the gap where bytes could be lost is closed by design. Everything lands in one file, in order.

It ends when your device goes quiet after a credit, or when it reaches one of two backstops: **32 windows** or **20 minutes**. There is no key that ends it early. Those exist so a device that never stops talking cannot leave the panel busy for ever, and the 20 minutes is usually the one you meet — decoding a window takes longer than recording it, so 20 minutes is about eight windows at 9600 baud rather than 32.

If it stops for either reason it says which, and it tells you how to carry on: **press Capture again**. A finished recording leaves you in the mode you chose — the title bar still reads `SERIAL DECODE - 32K CAPTURE` — so it is one press, and the bytes that follow go into a **new** numbered file rather than the one it was just writing.

No bytes are lost in the gap. Your device only transmits when it is credited, and no credit goes out until the next recording arms, so it is still waiting exactly where the run stopped. What you get is one conversation split across two files, in order, with nothing missing between them.

Three things your device has to get right:

- **Catch a 5.7 μ s pulse.** That is short: use an interrupt, an edge-triggered input or a hardware latch. A firmware polling a GPIO in a loop can miss it, and at 250000 baud 5.7 μ s is under 1.5 bit times. It is a pulse per window, not a level you can poll once.
- **Send no more than one window per pulse** — 8 192 or 32 768 bytes depending on the mode. The pulse carries no size information, so the limit has to be built into your firmware.
- **Go quiet when it has nothing to send.** That silence is what tells the app the transmission is over.

Honest limitation: the pulse itself is verified — +4.45 V, 5.72 μ s, 436 ns rise, one per arm, measured on a scope — and the loop that issues them is tested. The full round trip, with a device that actually waits for the pulse, has **not** been tested here: the signal generator on this bench stayed silent through a credit, and whether that is its trigger input or its burst arming has not been separated yet. The leading suspect is the width, since 5.72 μ s is the narrowest pulse this bench can present and the generator's own minimum is not specified anywhere. So the mechanism is sound by construction and you would be the first to close the loop. **Triggering, in both directions** in [REFERENCE.md](#) has it gate by gate.

How long a recording gets you

There are two different times here and it is worth keeping them apart:

- **Signal** — how much of the line ends up in the file.
- **Waiting** — how long you stand there before the buffer is full.

Below about 19200 baud they are the same. Above it they are not, because the meter writes about **100 000 readings a second** into its memory however fast the digitizer is told to sample. The samples are properly spaced and the record is complete; gathering them just takes longer than the stretch of line they represent, so you stand there longer than the signal you get.

That ceiling is a separate thing from the sample rate, and the two are easy to confuse. The **sample rate** decides how many samples land in each bit — a frame capture gets very nearly the rate it asks for, a recording's burst gets somewhat less, and that difference is what limits recording to 153600 baud (above). The **100 000 readings a second** is not about fidelity at all: it is how fast readings reach memory, and it decides only how long you wait.

Locked rate	Signal you get	Waiting for the full 32 kB
300 Bd	1092 s	the same
1200 Bd	273 s	the same
2400 Bd	136 s	the same
4800 Bd	68 s	the same
9600 Bd	34 s	the same
19200 Bd	17 s	the same
38400 Bd	8.5 s	about 17 s
57600 Bd	5.7 s	about 17 s
115200 Bd	2.8 s	about 18 s
153600 Bd	2.1 s	about 21 s

Every row is the same 32 768 bytes — what changes is how long that takes to go past. The 8 kB window is a quarter of each figure. Then add the decoding, which does not depend on the baud rate: about two minutes for 32 kB, half a minute for 8 kB.

You do have to fill it. There is no way to stop a recording early — see the TRIGGER key section above — so choose the window with the whole wait in mind.

153600 baud is the fastest rate that can record. Above that, FRAME only.

The reason is that a recording and a frame capture do not get the same sample rate out of the same request. A frame capture gets very nearly the 1 MS/s it asks for; a recording's burst acquisition loses about half a microsecond per sample, so the same request delivers about 662 kS/s. The floor is four samples per bit, and the crossover works out at **165563 baud** — so of the standard rates, 153600 is the last one that clears it, at 4.3 samples/bit.

Above it the app **refuses before you press**, on the note row, naming the figure a recording would actually get:

192000 Bd: a recording gets 3.4 samples/bit, under the 4 floor -- use FRAME

That figure is deliberately the recording's, not the header's. The SA/BIT cell will read about 5.2 at 192000 baud, which is true of the frame capture it was measured from — the two are different acquisitions, and the note says which one it means.

FRAME keeps working to the full 250000 baud, and usefully so: at 192000 baud a frame capture holds 384 bytes.

Reading the bytes

The hex view shows two groups of eight bytes and then the same bytes as text. The offset down the left is dimmed because it is there to navigate by, not to read. A byte that could not be recovered shows as --.

ERR 0 does not mean the bytes are definitely right. It counts broken stop bits and parity failures. Noise landing inside a data bit flips it with nothing to detect — which is why the note row matters more than the error count.

If you see **began mid-byte – the first N bytes are misaligned.** : on a line that never goes quiet, the capture starts wherever it starts, and if that is halfway through a byte the first few bytes are chopped up wrongly. They are shown rather than hidden, but **do not trust their values**. Everything after the first gap in the traffic is fine. On traffic with normal gaps between messages it costs **5 to 12 bytes**; on a line that never pauses it can be far more — **up to 165 bytes measured** on back-to-back text at 250000 baud, because there is no gap to resynchronise on. A device that pauses between messages never shows it at all.

Locking the baud rate, and the amber padlock

The app locks the rate for you after a good capture, which is why later captures are quicker. It only does this when it is confident: the rate landed on a standard one, enough clean frames came through, **FIT** was 0.9 or better, and the voltage levels were steady.

When it is not confident it leaves the padlock **amber** and offers you the **Lock Rate** button. The bytes are still shown — they just are not trusted enough to carry forward. Press **Lock Rate** to pin it anyway, or capture again.

Auto-lock never locks polarity. It is worked out fresh on every capture, on purpose: guessing it wrong once and then sticking to it is how you get a confident, completely wrong decode. (**Lock Detected** on the Options screen *does* copy the detected polarity into the fields along with everything else — but there you can see what it picked before you commit it.)

If the line changes rate, the app gives the lock up

A locked rate is a promise about the wire, and when the wire stops keeping it the lock is worse than useless — it turns every frame into an error and nothing gets better by pressing Capture again. So when **all three** of these hold, the app **drops the lock by itself** and re-detects from the same samples:

- a rate is locked;
- the bit time it measures on the wire disagrees with that rate;
- more than a quarter of the frames in the middle of the capture failed.

It says so on the note row, naming the rate it discarded — **19200 baud fit nothing -- unlocked, and this capture reads 38400** — and from there auto-lock pins the new rate on the usual terms. That is the same result as typing **0** into Options by hand, which is what you had to do before.

The middle condition is what protects a rate you typed deliberately. If the rate fits the wire and the *format* is wrong, every frame fails too — but the measurement agrees with your number, so your number is kept.

Options

SERIAL DECODE OPTIONS

End App

Baud Rate
0 = auto-detect

9.60000kBd

Data Bits
auto, or fix it

8

Parity
auto or fixed

None

Polarity
idle level

Auto

Trigger
capture start

Start bit

Rear BNC
trig in / flow-control out

Off

Auto Detect

Lock Detected

Apply

Cancel

Setting	What it does
Baud Rate	0 means auto-detect. Type a number to lock that rate; it is rounded to a whole baud.
Data Bits	Leave on Auto (7/8) . Only use Auto (any) if you know the device sends 9-bit words — it can make a damaged capture look tidier while being wrong.
Parity	Auto , or pin None / Even / Odd .
Polarity	Auto , Idle high (normal CMOS/TTL) or Idle low (inverted). Auto is reliable unless the line is almost never idle.

Each setting is independent — pinning one leaves the rest on auto.

Apply commits your changes and takes a capture. **Cancel** puts everything back to how it was when you opened the screen. **Auto Detect** forgets everything locked, takes a fresh capture, and locks whatever that finds — it is the quick way back when you have locked onto the wrong thing. **Lock Detected** copies the last capture's settings into the fields so you can look at them before committing.

There is no stop-bit setting. A second stop bit looks exactly like a bit of idle line, which the decoder already copes with, so a 2-stop device decodes correctly and is reported as 1 stop.

Advanced

Two settings you can ignore unless you need them:

- **Trigger** — what starts the acquisition. **Start bit** (the default) waits for the line to move. **Free run** grabs immediately. **Trigger key** waits for the front-panel TRIGGER key, as described under **The TRIGGER key** above. **If the thing you asked for never happens, the capture goes ahead anyway** after a few seconds and the note row names the one that did not arrive — **edge trigger**

unavailable; captured free-running, or front for the TRIGGER key. Those bytes are real but they are not lined up with the event you wanted.

- **Rear BNC** — lets an external trigger in on the rear connector, and FC Out sends the flow-control credit pulse out of EXT TRIG OUT. FC Out is what turns a recording into an unlimited one; see **More than one window: flow control** above.

An external pulse on the rear connector really does start a capture — verified on this hardware, and verified by the negative as well: take the pulse away and the capture refuses rather than quietly grabbing something, with the note row naming what was missing. Two things to know before you rely on it. The rear input is **OR'd with** whatever **Trigger** says rather than replacing it, so the first of the two to happen wins the capture — and on a line that is always transmitting that is the start bit, every time, which leaves your pulse no say in where the window opens. Point it at a line that is otherwise quiet, or pair it with **Trigger key** so that the only thing it competes with is your own finger. And under **Free run** the rear input is ignored altogether, because free run means do not wait for anything — the app says so rather than leaving you to notice: the note row reads **Rear BNC = Trig In is set but IGNORED**, and the status cell marks it by appending a **?** to **EXT TRIG IN**.

Trigger applies to FRAME captures. **A recording ignores it** and starts as soon as you press Capture — for a recording, the thing that decides when the device talks is the flow-control credit, not a trigger. **FC Out** is not confined that way — it pulses once per armed capture whichever mode you are in, a FRAME capture included. What a recording adds is the repetition: credit, record, decode, credit again.

Files on the USB key

Two different files, and they are not the same thing:

File	Written	Holds
/usb1/SerialFiles/bytesNNN.txt	automatically, on every capture	the running log — every capture's bytes as text and hex. NewLog starts a new number.
/usb1/SerialFiles/serial_NNN.txt	only when you press Save	a full report of what is on screen: settings, all the notes, the whole dump.

Everything goes in /usb1/SerialFiles, and the app creates that directory the first time it needs it — there is nothing to set up, and nothing lands in the root of your key. The panel still shows the bare filename rather than the path, because the directory never changes and the row is narrow.

A recording decodes into the log file, so the file holds the whole run while the panel shows only the tail of it. The note row says so when they differ.

Recording needs a USB key — the file is the point of it. Normal FRAME captures work without one; they just cannot log or Save.

When something looks wrong

What you see	What it means
line is idle (no transitions)	Nothing is crossing the threshold. Check your connection, and check the device is actually talking while you capture.

What you see	What it means
no clear logic levels	Too much noise, or the swing is too small. 0.33 V works on the bench, and 0.25 V did too; 60 mV is refused outright.
Wrong rate reported	The traffic may be genuinely ambiguous (below). Lock the rate if you know it.
Bytes wrong but ERR 0, with the rate locked	The device is not running at the rate you locked. The note row warns when the line disagrees by more than 4 %.
A baud rate that is not a round number	It is reporting what it measured, because the measurement was not within 2 % of a standard rate.

Some traffic is genuinely ambiguous, and no decoder can fix it. Eight 0x00 bytes at 9600 baud 7N1 make exactly the same waveform as four 0x08 bytes at 4800 baud 8N1. The app tells you a rival rate fits rather than pretending to know. Irregular gaps between bytes remove the ambiguity completely.

Two cases where automatic detection is known to fail are described under **Known failures of automatic rate detection**, below. Both are fixed by typing the rate in.

What it can cope with

Measured at 8 samples per bit. Past every one of these limits the app **reports** the problem rather than quietly getting it wrong, except where noted.

Condition	Works up to
Wrong clock, rate locked	±4 %. Beyond that a fast device corrupts bytes with nothing to show it
Wrong clock, auto-detect	±12 % and past — it measures instead of assuming
Jitter	±15 % of a bit; past ±20 % it can fail quietly
Noise	about 40 % of the logic swing — roughly 1.3 V on a 3.3 V line
Slow edges, long cables, weak pull-ups	filtering up to 0.7 bit times
Logic swing	0.33 V to 8 V — the limit is the swing, not the logic family
DC offset, logic low at or above 0 V	anything that keeps the line inside ±10 V
DC offset, logic low below 0 V	works, but see the note below — one band is not reliable

Rates from 300 to 250000 baud decode byte-exact on the bench, and so do odd ones — 900, 1500, 3600, 7200 and arbitrary values like 1379, 8123 and 104857. A 1 kB non-repeating payload is clean at every one of the eleven standard rates from 300 to 250000, and 30 consecutive captures at 2400 baud returned ~6 780 bytes with nothing flagged. A rate within 2 % of a standard one is reported as the standard one, because real devices use standard rates.

Tolerance follows samples per bit, not baud rate. A slow line is not automatically the safe one — read SA/BIT.

If you are writing the firmware at the other end, two things help: keep your edges within ±15 % of a bit time (that is ±15.6 µs at 9600 baud but only ±1.3 µs at 115200, so if an interrupt can delay your transmit loop, send slower), and avoid padding with 0xFF, which gives the rate detector only one narrow pulse per frame to measure. 0x00 padding is fine.

Known failures of automatic rate detection

Read this in proportion. Most serial decoders require you to tell them the baud rate. This one measures it, and gets it right on every standard rate and on arbitrary ones like 1379, 8123 and 104857. Two cases are known where the measurement goes wrong — **8 points in 860** on a full bench plan, and none of them an ordinary payload at a standard rate — and in both the remedy is to supply the rate, which is what other tools ask for as a matter of course.

Both are fixed by typing the rate into `Options ▶ Baud Rate` — verified, every frame decoded with zero errors once the rate was given. The numbers are below so you can judge the risk yourself.

Automatic detection has no clock to refer to. It works in two steps: it measures the gaps between transitions and finds the longest bit time that divides them all, then weighs that answer against the standard rates, because a real device almost always uses one.

Gaps of many different lengths pin the measurement down well; gaps of only a few lengths do not. But where detection fails it is usually not the measurement that is wrong — it is the second step. A different reading at a simple multiple of the measured rate can land on a standard value while the true rate does not, and that coincidence is allowed to outweigh a correct measurement.

Two checks stand between a correct measurement and its own multiples, and both apply only once the measurement itself lands on a standard rate — where the true rate is non-standard there is nothing to defend, which is why the failures below concentrate there.

- **Where the pulses fall.** At the true bit time some pulses span an odd number of bit cells; divide that bit time by an even number and every pulse spans an even count. So a halved, quartered or sixthed reading is recognisable from the pulse lengths alone.
- **How short the shortest pulse is.** Dividing by an *odd* number leaves the counts' parity exactly as it was, so thirds and fifths are invisible to the first check. What gives them away is that they claim a longer minimum. Ordinary traffic has single-bit runs — a ten-bit frame opens with a one-bit start bit — so a reading whose shortest run is three or five bit times is describing something else, and a reading past about two and a half is refused. **The cost is real and worth knowing:** a payload whose shortest genuine run is three bit times or more, such as `0x00` sent with a two-bit gap between bytes, cannot have a third-of-the-rate reading accepted. Such a payload reports a third of its true rate either way, so this refuses nothing that was working, but it is the shape to watch for.

1. A short pattern repeated over and over

What triggers it. Data built from `0x00`, `0xFF`, `0x55`, `0xAA`, walking bits, or one fixed string sent on a loop — link-training patterns and idle fill, rather than real messages. Such data yields only a handful of distinct gap lengths, and a simple fraction of the true bit time then explains those gaps just as well as the truth: half the bit time reads the same square wave as twice as many bits, with no framing error anywhere to give it away. Both readings describe the waveform, so nothing in the signal can rank them — and where one of them lands on a standard rate and the true rate does not, the standard one wins.

The measurement is not the problem. Traced on a failing case, the bit time was recovered as 8.3990 samples against a true 8.3987, every one of the smallest gaps sat at the true bit time, and the decode at that rate framed every byte without a single error. It was then overruled by its own half, because 3572 is not a standard rate and 7200 is.

What you see. The reported rate is a small whole-number multiple of the true one — two to five times — and always one of the standard rates. Expected 3572 and the panel says 7200, or expected 124000 and it says 250000. Usually the byte area also fills with error markers, because a wrong bit time shreds most

frames; in the measured cases 83 to 119 frames of a few hundred were flagged. Occasionally the frames survive and the bytes are simply wrong, which is the case worth knowing about.

There is no warning about the rate itself. Where two rates fit a waveform the panel normally names the runner-up under the reading, but not in this case — verified, the note is empty here. So the error markers are the whole of the signal, and where the frames happen to survive there is none. If a repeating test pattern reports a standard rate you did not configure, distrust it and lock the rate.

How rare, and where. Measured over **1 677 waveform-and-rate cells, each captured from 200 different points in the waveform — 335 400 captures.** The capture's opening point matters more than anything else here, because it decides how much of the first byte is missing and therefore how damaged the reading looks:

the true rate is	captures	reported wrong	
a standard baud rate	171 600	882	0.51 %
a non-standard rate	163 800	4 554	2.78 %

Every one of the 882 is an impaired signal, not merely a repeating pattern: three waveforms carrying 0.6 V of sinusoidal drift and one carrying heavy edge jitter. **No ordinary payload misreports at a standard rate at all** — the fox, the 1 kB text payload, twelve random payloads and the walking-bit patterns are clean at every standard rate from 300 to 250 000 baud, across all 200 opening points. That is 171 600 captures, and the split above is drawn from the same run.

A **LIN-style break field** — 13 or more bits held low, which is not valid 8N1 at all — is the hardest case the bench carries, and it reads correctly at 600, 19 200, 31 250, 38 400, 57 600 and 76 800 baud from every one of the 200 openings. This decoder does not claim to read LIN, and a break field is outside UART by construction; if you are probing a bus that begins frames with a long dominant field, the rate is worth setting by hand.

On the instrument the same plan gives 8 wrong rates in 860 capture-and-decode points, on three waveforms: the repeating single-value blocks, a LIN frame, and a heavy-jitter case.

One qualification, which is not this failure. Half of a standard rate is often *also* standard — 4800 is half of 9600 — and for perfectly regular traffic the two readings are the same waveform, as **Some traffic is genuinely ambiguous** above describes. That is not a misdetection and the panel names the rival rate when it happens; the failure in this section is the one where it does not.

The bench deliberately spends **half** its points on non-standard rates, which is far more than ordinary use, so the figure above overstates the exposure of a typical setup by roughly five times.

Besides locking the rate: capture real traffic rather than a test pattern, or let the bytes have irregular gaps — either removes the ambiguity.

2. Logic low sitting well below ground

What triggers it. A line whose low level is *negative* with respect to the probe's reference, with the whole signal still inside ± 10 V.

What you see. The rate comes back wrong, in the same multiple-of-the-truth way as above.

How rare. Most of the negative range is fine: measured clean at **-0.1, -0.25, -0.5, -1.0 and -3.0 V** on a 3.3 V swing. Around **-2 V** some payloads fail. It is not a threshold — **-3 V works where -2 V does not** — so it is a band rather than a limit. Across a full seeded sweep with amplitude and offset both varied, **12 of 1**

763 points were affected (**0.68 %**), and varying the amplitude alone over 0.5× to 1.6× caused **no failures at all**. Signals sitting entirely at or above 0 V are unaffected at any offset inside ± 10 V.

Besides locking the rate: reference the probe to the line's own low level rather than to earth.

Measured figures, the full tolerance envelope, press-by-press timings and the reasoning behind the design are in [REFERENCE.md](#).